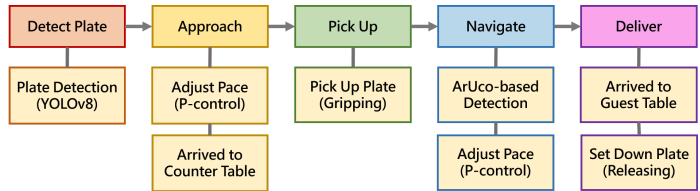


# Autonomous Dining Service Robot

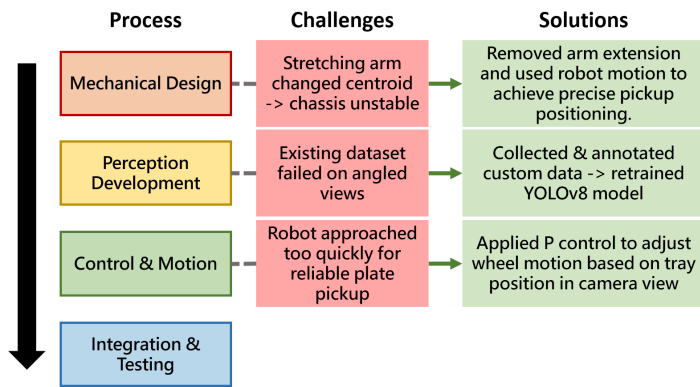
## Project Goals:

- Develop a small-scale autonomous dining service robot that detects, classifies empty and food-filled plates, picks up plates, and delivers them to designated tables in a simulated restaurant environment.

## Project Overview:



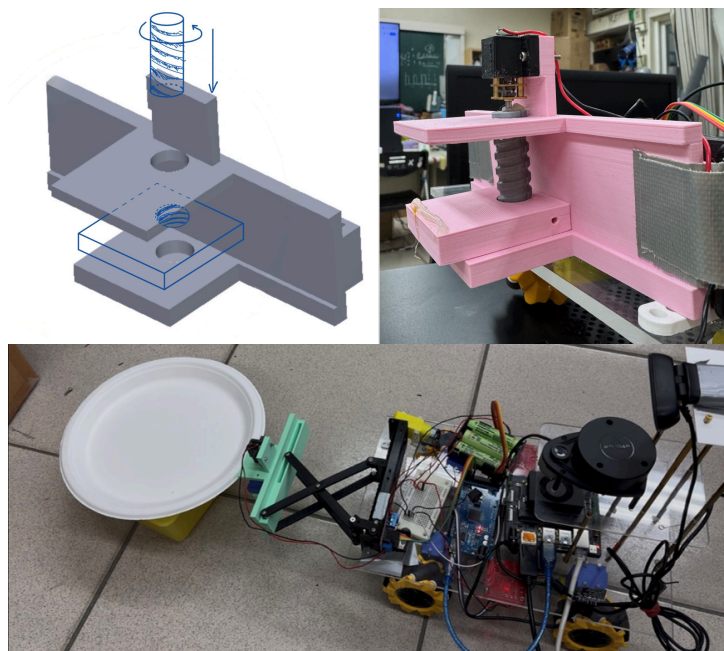
## Execution Process:



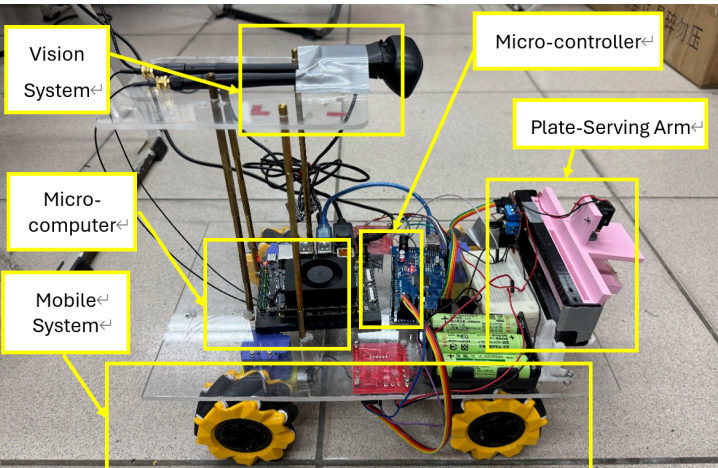
## Technical Implementation:

### Mechanism & Motion

Designed a lift-based gripper mechanism for plate pickup. The initial arm-extension design (the lower figure) shifted the robot's center of mass and caused chassis instability. Gripper design below:



## System Architecture:

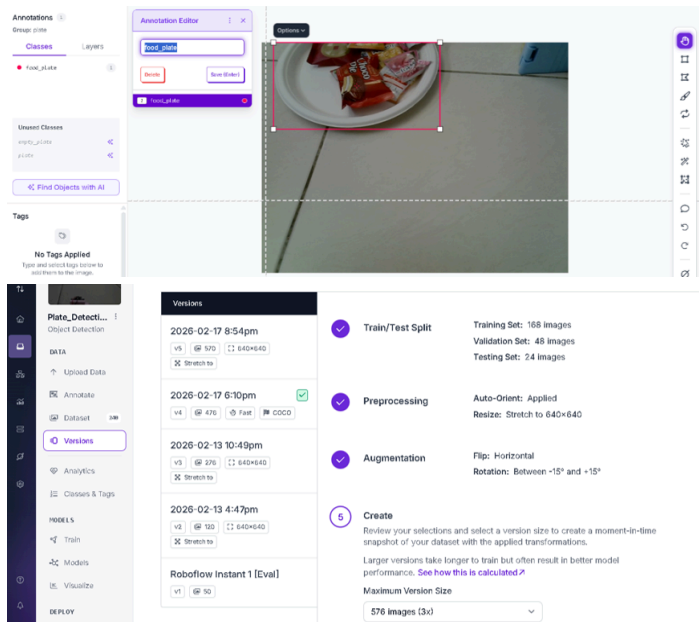


## Techniques & Tools:

- Mechanical Design:**  
SolidWorks / Lift-based gripper
- Perception Development:**  
YOLOv8 / OpenCV / Custom Dataset
- Control & Motion:**  
Mecanum Wheels / P-control / Camera View Feedback
- Integration & Testing:**  
ArUco Detection / Arduino Mega / NVIDIA Jetson Orin Nano

### Perception

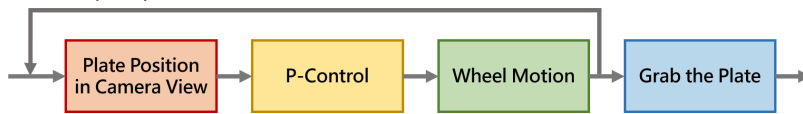
Existing datasets performed poorly on angled plate views since the camera was mounted at a fixed angle. I therefore collected and annotated a custom dataset and retrained a YOLOv8 model.



## Control & Integration

Used plate position in the camera view as feedback to adjust wheel motion during the final approach. ArUco markers were used to identify the target delivery location.

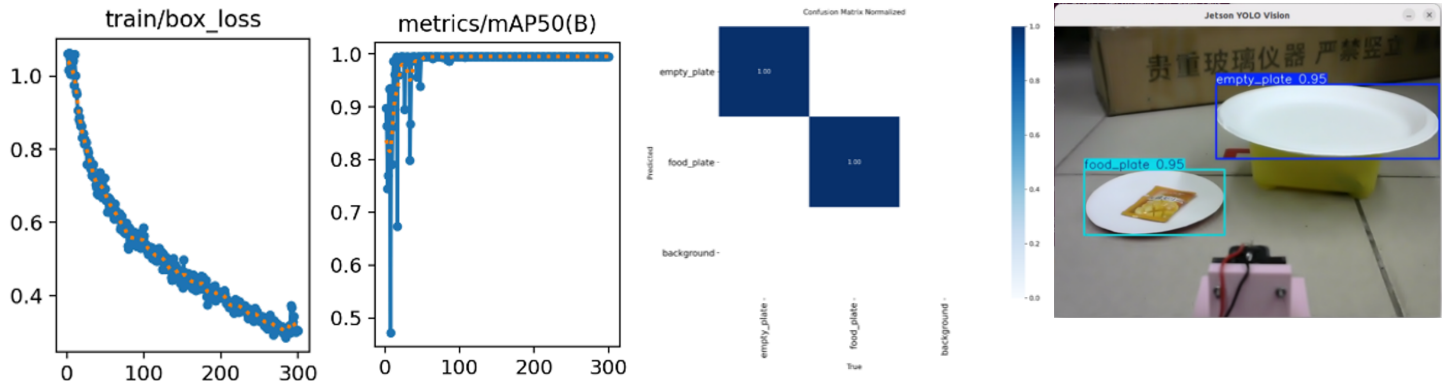
If plate position too close -> less wheel turns (slower)  
If plate position too far -> more wheel turns (faster)



## Result & Demonstration:

### ● Perception

The trained model detected and distinguished empty and food-filled plates in the tested scenes.



### ● System-Level Result & Demonstration

With the lift-based gripper design, trained YOLOv8 model, P-control adjustment, and ArUco marker detection, I successfully integrated perception, motion control, gripper positioning, and ArUco-based delivery into an autonomous serving workflow.

This is a video link regarding the autonomous dining service robot's demonstration.

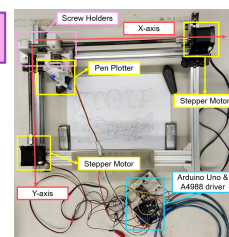
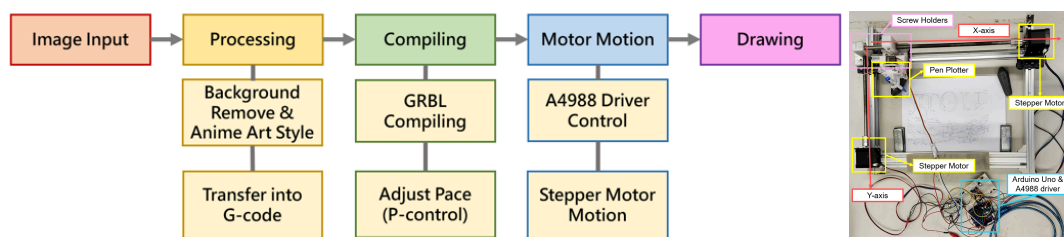
<https://youtu.be/sNF4TxTJK3E>



## Intelligent CNC Plotter Machine

### Project Overview:

A team project integrating mechanical design, image processing, G-code integration, and GRBL-based motion control for automated 2D drawing.



### My Contribution:

#### ● Mechanical Design

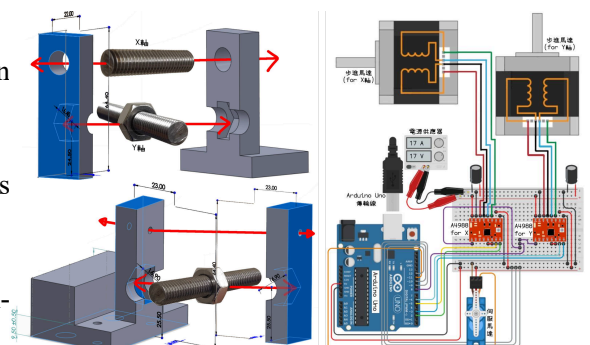
Designed two SolidWorks screw-support structures to maintain screw alignment and prevent binding during the motion.

#### ● Electronics

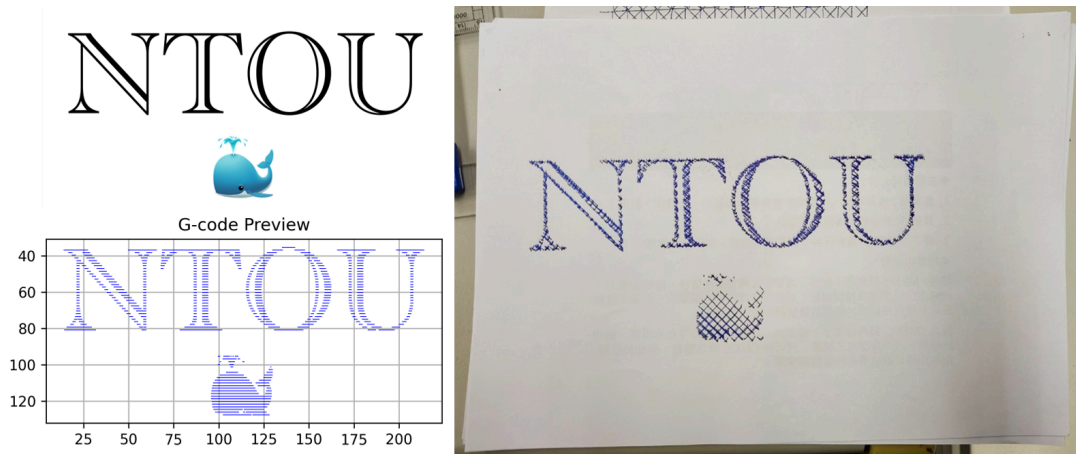
Wired the A4988 motor-driver circuit and integrated stepper motors with the GRBL control system.

#### ● Image Processing

Integrated OpenCV-based background removal into the team-developed Python GUI to simplify images before plotting.



## Result:



- **GitHub Repository:**

[https://github.com/Aynslielin/Intelligent-CNC-Plotter-Machine/blob/main/README\\_en.md](https://github.com/Aynslielin/Intelligent-CNC-Plotter-Machine/blob/main/README_en.md)

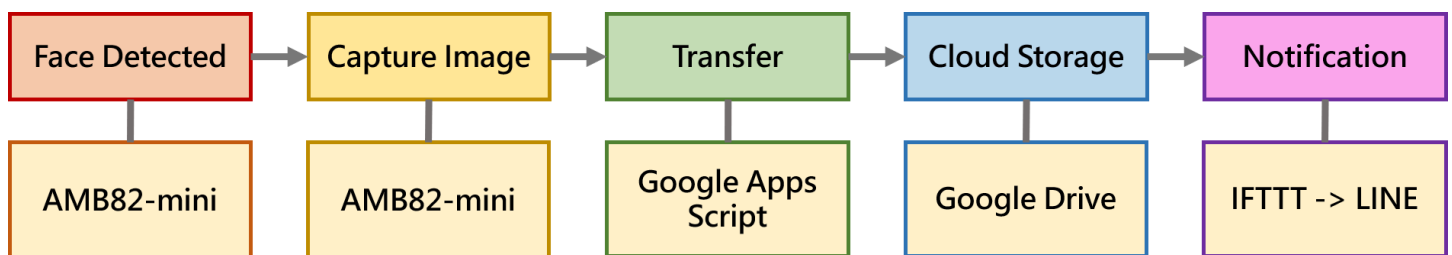
- **Demonstration Video List:**

[https://www.youtube.com/playlist?list=PLaIR\\_ZBq24bLc04j9EN7dDOz\\_QjDmvVbR](https://www.youtube.com/playlist?list=PLaIR_ZBq24bLc04j9EN7dDOz_QjDmvVbR)

## AI Doorbell

### Project Overview:

A team project that integrated image transfer from the AMB82-mini to Google Apps Script, then connected Google Drive and LINE through IFTTT to automate cloud storage and user notification.



### My Contribution:

Integrated image transfer from the AMB82-mini to Google Apps Script from Arduino IDE, and connected Google Drive with LINE through IFTTT to automate image storage and notifications.

## Result:

- **GitHub Repository**

<https://github.com/Aynslielin/Artificial-Intelligence-Doorbell>

- **Demonstration Video**

Video 1: <https://www.youtube.com/watch?v=YFg1xQITuxs>

Video 2: <https://www.youtube.com/watch?v=lumI-uAtkRU>

## Engineering Skills

### Mechanical Design:

SolidWorks / Gripper Mechanisms / Structural Alignment

### Embedded & Electronics:

Arduino / NVIDIA Jetson Orin Nano / Motor Drivers / Stepper Motors / Sensors

### Robotics & Control:

Mecanum Drive / Proportional Control / ArUco

### Programming & Scripting:

Python / C / G-code

### Computer Vision:

YOLOv8 / OpenCV / Data Annotation

### System Integration:

Hardware-Software Integration / Testing / Debugging